

# ARVIND SAH

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## PROFESSIONAL SUMMARY

Results-driven Data Engineer with 3+ years of experience designing scalable ETL pipelines, automating data workflows, and integrating diverse data sources across cloud platforms. Proficient in PySpark, Apache Spark, AWS, SQL, Snowflake, and Terraform. Strong backend engineering foundation with hands-on expertise in cross-database migrations (MySQL, PostgreSQL, Snowflake, Oracle, MSSQL, BigQuery), low-code ETL platform development, and distributed data processing. Passionate about transforming raw data into actionable insights and optimizing data infrastructure for performance, reliability, and scalability.

## PROFESSIONAL EXPERIENCE

### Data Engineer — Azminds Services Pvt. Ltd., Kathmandu Jan 2024 – Present

- Designed and developed DataFuseAI, a low-code ETL platform enabling non-technical users to build end-to-end data pipelines via a visual interface, supporting 15+ transformation types (Aggregate, Deduplicate, Filter, Join, Pivot, Unpivot, Window, Route, Split, Union, Derived, Rank).
- Built and integrated 20+ data source connectors: MySQL, MSSQL, PostgreSQL, Oracle, Snowflake, Redshift, MongoDB, Couchbase, Cassandra, BigQuery, AWS RDS, Azure Blob, S3, FTP, and SFTP.
- Engineered an Apache Livy-based execution engine enabling remote PySpark job submission for large-scale ETL operations without direct cluster access.
- Developed a SQL Dialect Conversion Platform translating SQL queries across MySQL, MSSQL, PostgreSQL, and Oracle using SqlGlott and custom regex-based transformation rules.
- Processed large-scale datasets using PySpark and Databricks, optimizing data transformation jobs for throughput and cost efficiency.
- Managed AWS infrastructure (EC2 clusters, load balancers, auto-scaling groups, Elastic IP); automated scaling policies using Terraform.
- Executed cross-database data migrations: MySQL → PostgreSQL, PostgreSQL → Snowflake, PostgreSQL → MySQL, MSSQL → BigQuery, Oracle → PostgreSQL via PySpark/JDBC with checkpoint/resume, row-count validation, and deduplication.
- Deployed Elasticsearch and Kibana for full-text search and log analytics in the YachtCloud platform.

### Backend Developer — Unlimited Remit Technology Dec 2022 – Dec 2023

- Built RESTful APIs for the Earn Sikka loyalty platform, enabling multi-channel Sikka distribution (phone number, QR code scanning, 6-digit code) with full redemption workflows.
- Designed and optimized relational database schemas in MySQL, PostgreSQL, and Oracle for efficient data storage and retrieval.
- Integrated third-party payment and identity verification services into the application backend.
- Collaborated cross-functionally with frontend developers and designers to deliver full-stack feature implementations using PHP (Laravel), MySQL, and Vue.js.

## EDUCATION

### M.Sc. in Computer Engineering — Specialization: Data Science and Analytics 2026 – Ongoing

*Pulchowk Campus, Tribhuvan University*

### B.E. in Computer Engineering 2017 – 2022

*Western Regional Campus, Pokhara*

Percentage: 70.58%

### Higher Secondary Education (NEB — Science) 2015 – 2017

*Arniko H.S.S, Biratnagar*

Percentage: 80%

### School Leaving Certificate (SLC) 2015

*Times School, Mirchaiya*

Percentage: 86.875%

## TECHNICAL SKILLS

**Languages:** Python, SQL, Scala, Groovy, JavaScript, PHP

Pandas

**Frameworks:** Grails/GORM, Laravel, Node.js, React, Vue.js,

**Big Data & ETL:** Apache Spark, PySpark, Apache Livy, Databricks,

dbt, SQLMesh

**Search & Logging:** Elasticsearch, Kibana, Logstash

**Databases:** PostgreSQL, MySQL, Oracle, MSSQL, Snowflake, Redshift, BigQuery, MongoDB, Couchbase, Cassandra

**Tools & Platforms:** Git, Databricks, Jupyter Notebook, PyCharm, IntelliJ IDEA, VS Code, Postman, n8n

**Cloud & DevOps:** AWS (EC2, S3, RDS, Lambda, SNS, Route 53, IAM), Terraform, Docker, Docker Compose, Nginx

## PROJECTS

### DataFuseAI — Low-Code ETL Platform —

2024 – Ongoing

- Built a low-code data integration platform supporting visual pipeline creation for ETL operations without writing code.
- Implemented 3 processing engines: Databricks, Apache Livy, and local PySpark execution.
- Supported 15+ transformation nodes: Aggregate, Deduplicate, Filter, Join, Pivot, Unpivot, Split, Union, Route, Derived, Rank, Window.
- Integrated 20+ data sources: RDBMS, cloud warehouses (Snowflake, Redshift, BigQuery), NoSQL (MongoDB, Cassandra, Couchbase), object storage (S3, Azure Blob), and FTP/SFTP.
- Built job scheduling, run history, and data profiling modules for pipeline observability.

**Tech Stack:** *Python, PySpark, Apache Livy, Databricks, AWS, PostgreSQL, Docker*

### Yacht Cloud — Charter Management Web Application —

2024 – Ongoing

- Built yacht management, booking, collaboration, and crew profile modules using Grails/GORM and Spring Security.
- Integrated Elasticsearch for full-text yacht search and AI-powered chatbot query resolution (offers, availability, booking status).
- Implemented S3 presigned URL uploads, gallery management, payment tracking with installment support, and expense management.
- Engineered async Playwright-based multi-site yacht charter scraper deployed on AWS Lambda (Docker), processing 1000s of listings with OpenAI Batch API normalization.
- Packages: Anchor (3 yachts), Harbour (5 yachts), Regatta (10 yachts), Admiral (unlimited).

**Tech Stack:** *Grails, Groovy, PostgreSQL, Elasticsearch, AWS S3/Lambda, PySpark, OpenAI AP*

### Text-to-SQL Engine — Natural Language Query Generation —

2025

- Built a natural-language-to-SQL engine that converts plain-English questions into validated, dialect-correct SQL using a locally hosted open-source LLM (Ollama + DeepSeek-Coder-v2 16B), with no query data leaving the environment.
- Implemented a 3-layer validation pipeline with self-correcting retries: a read-only keyword guard blocking 20 write/DDL operations, sqlglot AST parsing per dialect, and live EXPLAIN validation via SQLAlchemy — failed SQL and its error are fed back to the model for correction (up to 3 attempts).
- Supported 14 SQL dialects across 26 database drivers, including managed AWS RDS/Aurora and Azure variants, with per-engine EXPLAIN / SET PARSEONLY / EXPLAIN PLAN FOR routing.
- Engineered depth-limited schema extraction that walks foreign-key relationships from a chosen table instead of sending the whole database to the model.
- Designed a two-tier schema and session store: Redis as hot cache with TTL, PostgreSQL as durable backup that automatically repopulates Redis on cache miss or restart.
- Added multi-turn session context so follow-ups ("now only include last month") resolve against previous turns.

**Tech Stack:** *Python, Ollama, DeepSeek-Coder-v2, SqlGlott, SQLAlchemy, Redis, PostgreSQL, Docker*

### Blogs Automation Flow — n8n Workflow Automation —

2025

- Built an end-to-end n8n workflow that ingests incoming blog content via Gmail, deduplicates against a processing log, and extracts/sanitizes raw text for downstream processing.
- Integrated OpenAI (GPT) to generate structured blog post content from sanitized input, parsing the LLM output into validated JSON.
- Orchestrated REST API calls (login, blog creation, manager/invitee lookup) to publish blog content and notify stakeholders automatically via Gmail.
- Implemented state tracking across the pipeline (Processing → Processed/Error) using label-based Gmail folders to prevent duplicate runs and support failure recovery.

**Tech Stack:** *n8n, Gmail API, OpenAI API, REST APIs, JSON*

### Data Migration Pipelines — Cross-Database ETL —

2024 – 2025

- MySQL → PostgreSQL: Migrated large-scale relational datasets using PySpark JDBC connectors with schema mapping, type casting, and incremental load strategies.
- PostgreSQL → Snowflake: Built automated pipelines using PySpark and the Snowflake Spark connector, leveraging

staged S3 uploads for efficient bulk loading.

- PostgreSQL → MySQL: Implemented cross-RDBMS migration with data validation, null handling, and schema reconciliation.
- MSSQL → BigQuery: Designed CTE-based transformation chains (11 stages) followed by PySpark-to-BigQuery writes via GCS staging bucket.
- Oracle → PostgreSQL: Resolved data type compatibility issues (CLOB, NUMBER precision) and schema filtering during full-load migrations.
- Implemented checkpoint/resume support, row count validation, and duplicate detection across all migration pipelines.

**Tech Stack:** *PySpark, JDBC, Snowflake, BigQuery, AWS S3, GCS, PostgreSQL, MySQL, MSSQL, Oracle*

### SQL Dialect Conversion Platform — Query Translation Tool —

2024

- Developed a SQL dialect translator supporting cross-dialect query conversion across MySQL, MSSQL, PostgreSQL, and Oracle.
- Used the SqlGlott library combined with custom regex-based transformation rules for accurate syntax mapping.

**Tech Stack:** *Python, SqlGlott, Regular Expressions*

### Earn Sikka — Loyalty Web Application —

2022 – 2023

- Built a loyalty rewards platform allowing users to earn and redeem Sikka points via phone number, QR code, and 6-digit code channels.
- Designed RESTful APIs, relational database schema, and third-party service integrations.

**Tech Stack:** *PHP (Laravel), MySQL, Vue.js*

### CERTIFICATIONS & ACHIEVEMENTS

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- Coursera — Introduction to Data Engineering | [LINK](#)
- Attended Bootcamp: Data Science vs Data Engineering — Practical Distinctions and Career Paths | [LINK](#)